NUS School of Computing

FT5007 Capstone

Month < July 2026 > Report by Individual Team Member

Project Title: Agentic AI Chatbot for Consumer Banking

Reporting Period: July 2026

For each team member, please describe your achievements, upcoming plans (to-dos), challenges, and plans to overcome challenges in this period.

Student Name: Wu Qianzong

Achievement during the month:

- **Frontend:**
    - **UI:** implemented with basic conversational functions, support chat history viewing, user query input and submit, mock user login and logout, quick chips for follow-up questions.
- **Backend:**
    - **Workflow pipeline:** implemented a workflow using langGraph server to support the internal process from user input received at backend to response returned to frontend:
        - **Documents:** End-to-end process document v3.75 (End-to-End Process Specification), Mermaid workflow chart v3.75 (https://mermaid.ai/app/projects/907e3484-b8c8-4800-a5c8-dbc526f4b6b4/diagrams/20d927f9-dbb5-4728-9333-8e74e2459d25/version/v0.1/edit)
        - **Functions implemented:** Input pre-check, Intent classification, Knowledge blueprint, Knowledge retrieval, Knowledge sufficiency check, Resolution planner, response generator, deterministic gate post-check, precision substitution, cross-turn confirmation.
        - **Functions to be implemented next month:** semantic gate pre-check, semantic gate post-check, user escalation, deep-link, personal data lookup (out of scope but tentative for idea demonstration)
    - **RAG pipeline:**
        - **Documents:** RAG specification v0.4 (RAG Specification)

- **Data preparation:**
    - **Raw data:** Extracted SG Trustbank FAQ data from its official website
    - **Data synthesize:** For proprietary reasons, raw data cannot be directly used. Used LLM to generate synthesized FAQ data referencing raw data.
    - **Domain and intent mapping:** Use LLM to classify Synthesized FAQ by domain and intent(s) and mapping to project required domain and intent with a little expansion to for retrieval accuracy.
    - **Precision substitution:** For number accuracy, numbers in the synthesized data are replaced by a specific placeholder and the mappings of placeholder and number are stored in a relational database for future precision substitution. The original data without placeholders are preserved for vectorization for more accurate cosine similarity match.
- **RAG knowledge retrieval:**
    - **Chunking:** Chunking logic is optimized for searching. Child chunks for vectorization are designed into two types question child chunk and answer child chunk. For large answers, answer child chunk will be split into smaller ones with 150 characters overlapped for semantic continuation. Additionally, question will be prepended to each answer child chunk to increase the score of cosine similarity. And for answers of multiple intents, it will be duplicated before injection to vector db. Synthesized data has been processed according to the above rules for easy vector db injection.

To-dos in next month:

This month the whole objective is to achieve end-to-end basic conversational capability for one intent (account opening) and it is achieved. Next month the focus is to expand the knowledge base to all the FAQ data and add functions that make this chatbot suitable for consumer banking. Specific functions to build are:

Key Implementation Tasks:

1. Safety pipeline

- **Semantic gate:** Pre-check to identify malicious intent, non-compliant queries (Inbound semantic check); Post-check for relevance, groundedness and compliance
- **Deterministic gate:** Post-check for illegal characters, sensitive words, illegal placeholder or links

2. Workflow pipeline: User escalation

- case summary generation and escalation UI
- Hard escalation route
- Soft escalation route

3. RAG pipeline: full knowledge base

- vectorize and inject full knowledge base from synthesized data
- knowledge retrieval planning and search

4. Workflow pipeline: Memory:

- automatic summarize current session to shorten sliding window of chat history
- Summarize session snapshot at session end or timeout and store in database for next initialization greeting to recall last session data

4. Workflow pipeline: Tool calling

- Links to bank existing page (whitelisted)
- Personal data lookup (Dynamic personal data placeholder), although not in the MVP scope discussed in last interim report, but I'd like to implement a personal profile lookup to demonstrate the idea

Challenges Faced

1. **Prompt Engineering:** Initial behavior-based constraints failed to prevent LLM hallucinations where the bot acted as bank staff. Rules could not exhaustively cover all non-deterministic interactions, leading to unpredictable behaviors.

2. **Data Structure Design:** Standard flat chunking caused vector dilution and cross-domain confusion, resulting in poor search accuracy and low cosine similarity scores.

Plans to Overcome Challenges

1. **Principle-Based, Role-based Prompting:** Replaced fragile behavior-based rules with a role-based approach, instructing the bot to act as a "Strict Information Extraction Engine." The model is now restricted to grounding responses solely in the provided FAQ context, preventing its tendency to be overly helpful.

2. **Optimized RAG Topology:** Implemented a parent-child chunking structure to improve accuracy:

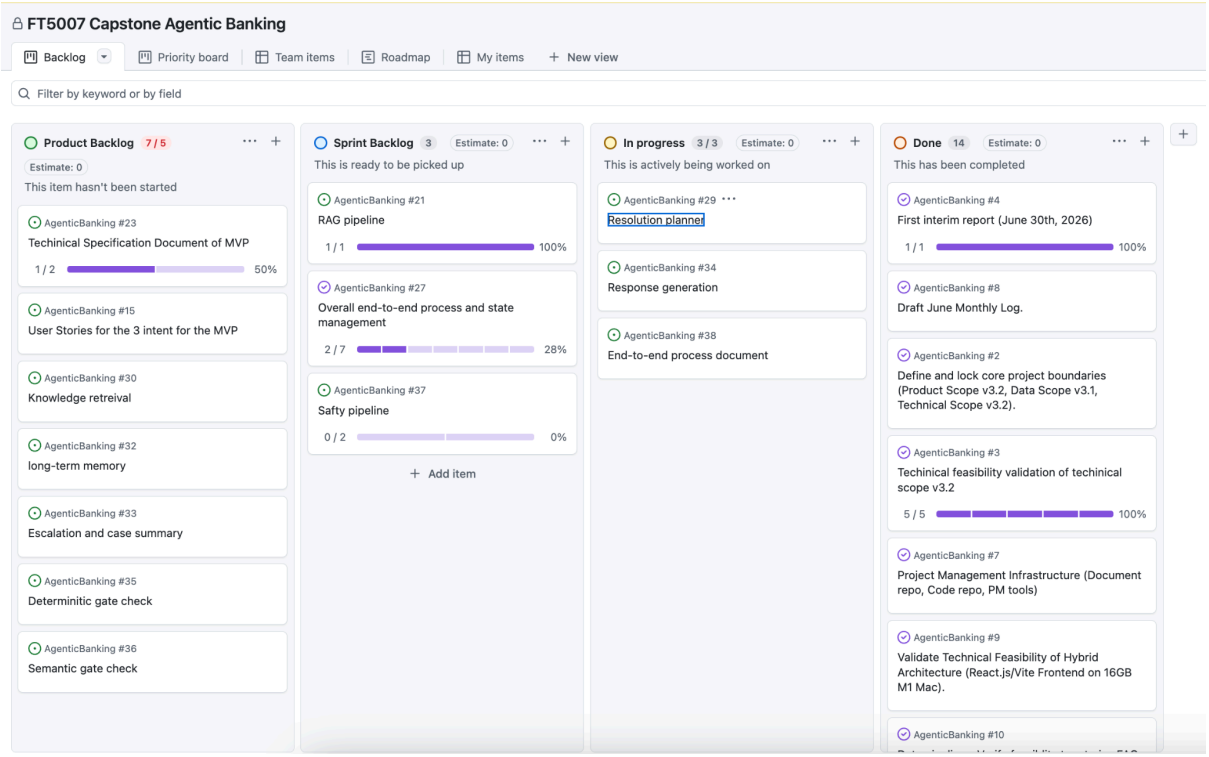
    - **Semantic Preservation:** Prepended questions to answer child chunks and used sliding overlaps for large answers to ensure semantic continuity.

    - **Domain anchored:** Duplicated chunks for multiple intents and utilized deterministic filtering to isolate domains, eliminating cross-domain noise and increasing cosine similarity scores.

Appendix:

Project Backlog:

https://github.com/users/wuqianzong/projects/2/views/1



Version control:

